

HINDUSTAN PETROLEUM CORPORATION LIMITED

(A Government of India Enterprise)

QUALITY CONTROL LABORATORY

MAZGAON TERMINAL

LUBE PLANT, HAY BUNDER ROAD, MAZGAON

MUMBAI,400033

LABORATORY TEST REPORT**HP NO. 1 10W 40 (210 L DRUM)(2578241)**

Laboratory Serial No.	PP-437	Sample Drawn By	QC OFFICER
JDE Serial No	1045159	Date Sample Drawn	24/04/21
Batch No / Lot No	M1-21-D-500	Date Received In Lab	24/04/21
Work Order No	833339	Date Tested	24/04/21
Tank /Kettle/Location	WHS	Particular of Sample	DRUM
Plant Code/ Name		Sampling Plan	IS 1447

Test Line Number	Test Description/ Characteristics	Units	Test Method	Specifications		Result Value
				Min	Max	
9	HP NO.1 10W 40 SULFATED ASH	%Wt	IS1448/ASTM/IP	1.22	1.54	1.33
1	HP No. 1 10W 40 APPEARANCE	.	VISUAL	B&C	B&C	B&C
8	HP No. 1 10W 40 CCS @ -25	Centepoise	P 108 IP 350	-	7000	5208
2	HP No. 1 10W 40 DENSITY	gm/cc	P 16 IP 180 D 4052	-	-	.857
10	HP No. 1 10W 40 FLASH PT	°C	P 69 IP 36	210	-	226
6	HP NO.1-TBN	mg KOH/g	ASTM/IP	9.4	-	10.3
7	HP NO.1 10W 40 ZINC	%Wt	D 6443	.107	.131	.117
3	HP No. 1 10W 40 KV @ 100°C	cSt	D 7042	14.0	15.0	14.4
5	HP No. 1 10W 40 POUR POINT	°C	P 10 IP 15	-	-27	-27
4	HP No. 1 10W 40 VIS INDEX	.	P 56 IP 226	120	-	155

Remarks : Sample meets the specification in above test parameter.

Note:

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Reviewed and Authorized by:
SHANBHAG G W(30044180)

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MUMBAI,400033

LABORATORY TEST REPORT**HP MILJUTE 460 DRUM (210 L)(1370241)**

Laboratory Serial No.	FB 871	Sample Drawn By	OPERATION OFFICER
JDE Serial No	1164500	Date Sample Drawn	28/07/22
Batch No / Lot No	M2-22-G-061	Date Received In Lab	28/07/22
Work Order No	965843	Date Tested	28/07/22
Tank /Kettle/Location	WHS2	Particular of Sample	DRUM
Plant Code/ Name		Sampling Plan	IS 1447

Test Line Number	Test Description/ Characteristics	Units	Test Method	Specifications		Result Value
				Min	Max	
1	APPEARANCE	.	VISUAL	B&C	B&C	B&C
2	ASTM COLOUR	ASTM	P12 D1500	-	6.0	3.5
3	KV @ 40°C	cSt	P25 D445 D7042	420.00	500.00	478.30
4	DENSITY@29.5°C	gm/cc	P16 D1298 D4052	-	-	.8957
5	FLASH POINT	°C	P69 D92	240	-	318
6	POUR POINT °C	°C	P10 D97	-	0	-6

Remarks : Sample meets the specification in above tests.

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Reviewed and Authorized by:

PAKALE SANTOSH BHIKAJI(38031160)

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LABORATORY TEST REPORT**SEETUL N 68 (P) (1X210L)(2317241)**

Laboratory Serial No.	PP 945	Sample Drawn By	QC OFFICER
JDE Serial No	1188138	Date Sample Drawn	10/10/22
Batch No / Lot No	M1-22-K-484	Date Received In Lab	10/10/22
Work Order No	985437	Date Tested	10/10/22
Tank /Kettle/Location	WHS	Particular of Sample	DRUM
Plant Code/ Name		Sampling Plan	IS 1447

Test Line Number	Test Description/ Characteristics	Units	Test Method	Specifications		Result Value
				Min	Max	
1	SEETUL N 68 (P) APPEARANCE	.	0	B&C	B&C	B&C
2	SEETUL N 68 (P)COPSTRIPCOR@100	.	P 15 D 130 IP 154	-	1	1
3	SEETUL N 68 (P) DENSITY @ 29.5	gm/cc	P 16 D 1298 IP 160	-	-	.8533
4	SEETUL N 68 (P) DIELETRIC STR	.	D 877 IP 295	20	-	26
5	SEETUL N 68 FLASH POINT	°C	P 69 D 92 IP 36	210	-	264
6	SEETUL N 68 (P) KV @ 40	cSt	P 25 D 445 IP 71	65.00	73.00	68.94
7	SEETUL N 68 (P) MOISTURE	Parts Per Million	D 1744	-	100	0
8	SEETUL N (P) NEUT NO.	mg KOH/g	P 2 D 974 IP 139	-	.05	.03
9	SEETUL N 68 (P) POUR POINT	deg.C	P 10 D 97 IP 15	-	-27	-30
10	SEETUL N 68 (P) VI	.		100	-	113

Remarks : Sample meets the specification in above tests.

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LABORATORY TEST REPORT**RUSTOP 274 (210 L)(0667241)**

Laboratory Serial No.	PP-1050	Sample Drawn By	QC OFFICER
JDE Serial No	1195792	Date Sample Drawn	12/11/22
Batch No / Lot No	M1-22-L-762	Date Received In Lab	12/11/22
Work Order No	995003	Date Tested	12/11/22
Tank /Kettle/Location	WHS	Particular of Sample	DRUM
Plant Code/ Name	14546	Mazgaon TML Lubes-(STP 0101)	Sampling Plan
			IS 1447

Test Line Number	Test Description/ Characteristics	Units	Test Method	Specifications		Result Value
				Min	Max	
1	RUSTOP 274 ACTION POLYBAG@45°	.	SKF-METHOD	N	N	N
7	RUSTOP 274 APPEARANCE	.	VISUAL	C	C	C
2	RUSTOP 274 DENSITY@29.5	gm/cc	P 16 D 1298 IP 160 D 4052	-	-	.872
4	RUSTOP 274 FILM THICKNESS	Microns	SKF-METHOD	8	-	10
3	RUSTOP 274 K V @ 40	cSt	P 25 D 7042 D 445 IP 71	18.00	28.00	26.30
6	RUSTOP 274 WATER	%Wt	P 40 D 95 IP 74	-	.1	.1
5	RUSTOP274WTAERDISPLACEMENTTEST	.	SKF-METHOD	P	P	P

Remarks : SAMPLE MEETS THE SPECIFICATION IN ALL ABOVE TEST PARAMETERS.

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Reviewed and Authorized by:

WARANG CHANDRAKANT PRABHAKAR(30074520)

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LABORATORY TEST REPORT**PARTHAN EP 320 (210 L)(0883241)**

Laboratory Serial No.	FB-1332	Sample Drawn By	OPN OFFICER
JDE Serial No	1197052	Date Sample Drawn	17/11/22
Batch No / Lot No	M2-22-L-749	Date Received In Lab	17/11/22
Work Order No	996260	Date Tested	17/11/22
Tank /Kettle/Location	WHS2	Particular of Sample	DRUM
Plant Code/ Name		Sampling Plan	IS 1447

Test Line Number	Test Description/ Characteristics	Units	Test Method	Specifications		Result Value
				Min	Max	
1	APPEARANCE	.	VISUAL	C	C	C
2	ASH	%Wt	IS 1448(P4), ASTM D482	-	.100	.040
3	DENSITY@29.5°C	gm/cc	P16,ASTMD4052, D1298, D7042	-	-	.8934
4	FLASH POINT°C,COC	°C	IS1448(P69),ASTMD92	220	-	254
6	FOAMING TEST SEQ1	Milliliter	IS1448(P67), ASTM D892		100/0	/0
7	FOAMING TEST SEQ2	Milliliter	IS1448(P67), ASTM D892	-	50/0	0/0
8	FOAMING TEST SEQ3	Milliliter	IS1448(P67), ASTM D892	-	100/0	0/0
9	RUST TEST A/B	.	IS1448(P96),ASTMD665	P	P	P
12	PARTHANEP 320 VI	.	P 56 D 2270 IP 226	90	-	97
10	KINEMATIC VISCOSITY @40°C	cSt	IS1448(P25),ASTMD445, D7042	300.0	340.0	338.1
5	EMULSION TEST@82°C/30MIN	Milliliter	IS1448(P91), ASTM D1401	40-40-0	40-40-0	40-40-0
11	PHOSPHORUS	%Wt	IS1448(P101),ASTMD1091, D6443	.0098	.0145	.0103

Remarks : SAMPLE MEETS THE SPECIFICATION IN ALL ABOVE TEST PARAMETERS.

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LABORATORY TEST REPORT**WAYLUBE 220 (210 L)(0402241)**

Laboratory Serial No.	PP - 1090	Sample Drawn By	QC OFFICER
JDE Serial No	1198568	Date Sample Drawn	23/11/22
Batch No / Lot No	M1-22-L-864	Date Received In Lab	23/11/22
Work Order No	997791	Date Tested	24/11/22
Tank /Kettle/Location	WHS	Particular of Sample	DRUM
Plant Code/ Name		Sampling Plan	IS 1447

Test Line Number	Test Description/ Characteristics	Units	Test Method	Specifications		Result Value
				Min	Max	
1	WAYLUB 220 PHOSPHORUS	%	P 33 D 129	.010	.050	.018
2	WAYLUBE 220 APPEARANCE	.	VISUAL	C	C	C
3	WAYLUBE 220 CORRCuSTRIP@100	Hour	P 15 D 130 IP 154	-	1	1
4	WAYLUBE 220 DENSITY@29.5	gm/cc	P 16 D 1298 IP 160	-	-	.889
5	WAYLUBE 220 KV @ 40	cSt	P 25 D 445 IP 71	200	220	215
6	WAYLUBE220NEUTRALISATIONNUMBER	mg KOH/g	P 2 D 974 IP 177	-	1.7	.6
7	WAYLUBE220SULPHURCONTENT(ADD)	%Wt	P 33 D 129 IP 61	.11	.55	.12

Remarks : Sample meets the specification in above tests.

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Reviewed and Authorized by:
DALVI P Y(30044160)

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LABORATORY TEST REPORT**HP MILJUTE 460 DRUM (210 L)(1370241)**

Laboratory Serial No.	FB 1474	Sample Drawn By	OPNS OFFICER
JDE Serial No	1206800	Date Sample Drawn	16/12/22
Batch No / Lot No	M2-22-M-832	Date Received In Lab	16/12/22
Work Order No	1006521	Date Tested	16/12/22
Tank /Kettle/Location	WHS2	Particular of Sample	DRUM
Plant Code/ Name		Sampling Plan	IS 1447

Test Line Number	Test Description/ Characteristics	Units	Test Method	Specifications		Result Value
				Min	Max	
7	APPEARANCE	.	VISUAL	B&C	B&C	B&C
7	ASTM COLOUR	ASTM	P12 D1500	-	6.0	4.0
7	KV @ 40°C	cSt	P25 D445 D7042	420.00	500.00	491.20
7	DENSITY@29.5°C	gm/cc	P16 D1298 D4052	-	-	.8942
7	FLASH POINT	°C	P69 D92	240	-	322
7	POUR POINT °C	°C	P10 D97	-	0	-6
7	RUSTOP 283 CORRCUSTRIIP@100	.	P 15 D 130 IP 154	-	1	1

Remarks : SAMPLE MEETS THE SPECIFICATION IN ABOVE TESTS

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Reviewed and Authorized by:

AGARWALLA PRIYANKA(31960370)

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LABORATORY TEST REPORT**HP MILJUTE 460 DRUM (210 L)(1370241)**

Laboratory Serial No.	FB - 1679	Sample Drawn By	OPERATION OFFICER
JDE Serial No	1215416	Date Sample Drawn	21/01/23
Batch No / Lot No	M2-23-A-151	Date Received In Lab	21/01/23
Work Order No	1015121	Date Tested	23/01/23
Tank /Kettle/Location	WHS2	Particular of Sample	DRUM
Plant Code/ Name		Sampling Plan	IS 1447

Test Line Number	Test Description/ Characteristics	Units	Test Method	Specifications		Result Value
				Min	Max	
1	APPEARANCE	.	VISUAL	B&C	B&C	B&C
2	KV @ 40°C	cSt	P25 D445 D7042	420.00	500.00	480.30
3	DENSITY@29.5°C	gm/cc	P16 D1298 D4052	-	-	.8923
4	FLASH POINT	°C	P69 D92	240	-	312

Remarks : Sample meets the specification in above tests.

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LABORATORY TEST REPORT**HP MILCY POWER 15W 40 (210L)(0818241)**

Laboratory Serial No.	FB 1703	Sample Drawn By	QC OFFICER
JDE Serial No	1216694	Date Sample Drawn	28/01/23
Batch No / Lot No	M1-23-A-481	Date Received In Lab	28/01/23
Work Order No	1017115	Date Tested	28/01/23
Tank /Kettle/Location	WHS	Particular of Sample	DRUM
Plant Code/ Name	14546	Mazgaon TML Lubes-(STP 0101)	Sampling Plan
			IS 1447

Test Line Number	Test Description/ Characteristics	Units	Test Method	Specifications		Result Value
				Min	Max	
6	TBN	mg KOH/g	P86 D2896	9.00	-	11.63
1	APPEARANCE	.	VISUAL	B&C	B&C	B&C
11	CCS @ -20°C	Centi Poise	P108 D5293	-	7000	6842
3	DENSITY@29.5°C	gm/cc	P16 D1298 D4052 D7042	-	-	.8839
4	FLASHPOINT	°C	P69 D92	190	-	226
8	KV @ 100°C	cSt	P25 D445 D7042	14.00	15.20	14.41
7	KV @ 40°C	cSt	P25 D445 D7042	-	-	106.0
5	POUR POINT	°C	P 10 D 97	-	-21	-30
2	SULFATEDASH	%Wt	P 4 D 874	1.190	1.420	1.200
9	VISCOSITY INDEX (VI)	.	P56 D2270	120	-	139
10	ZINC	%Wt	D6443 EDTA	.11	.13	.12
15	CALCIUM	%Wt	ASTMD6443,EDTA	.391	.478	.422
15	SULFATED ASH	%Wt	P 4 D 874 IP 163	1.50	1.83	1.77
15	TBN	mg KOH/g	P86 D2896	11.3	-	11.6
15	ZINC	%Wt	ASTMD6443,EDTA	.112	.137	.118

Remarks : Sample meets the specification in above tests.

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Reviewed and Authorized by:
PARVEEN RUBANA(31989940)

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LABORATORY TEST REPORT**PARTHAN EP - 680 (210 L)(0712241)**

Laboratory Serial No.	FB-1844	Sample Drawn By	OPERATION OFFICER
JDE Serial No	1224824	Date Sample Drawn	25/02/23
Batch No / Lot No	M2-23-B-356	Date Received In Lab	25/02/23
Work Order No	1027347	Date Tested	25/02/23
Tank /Kettle/Location	WHS2	Particular of Sample	DRUM
Plant Code/ Name		Sampling Plan	IS 1447

Test Line Number	Test Description/ Characteristics	Units	Test Method	Specifications		Result Value
				Min	Max	
2	PARTHAN EP 680 APPEARANCE	Visual	VISUAL	C	C	C
3	PARTHAN EP 680 ASH	%Wt	P 4	-	.10	.05
4	PARTHAN EP 680 DENSITY@29.5	KG / M3	P 16 D 1298 IP 180	-	-	.895
6	PARTHAN EP 680 EMULSION 60 M	Milliliter		-	0	0
5	PARTHAN EP 680 FLASH POINT	°C	P 69 D 92 IP 36	230	-	270
7	PARTHAN EP 680 FOAM TEST SEQ1	.	P 67 D 892 IP 136	-	100/0	0/0
8	PARTHAN EP 680 FOAM TEST SEQ2	.	P 67 D 892 IP 136	-	50/0	20/0
9	PARTHAN EP 680 FOAM TEST SEQ3	.	P 67 D 892 IP 136		100/0	0/0
1	PHOSPHOROUS	%Wt	P101 D1091 D6443	.0098	.0145	.0113

Remarks : SAMPLE MEETS THE SPECIFICATION IN ALL ABOVE TEST PARAMETERS.

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LABORATORY TEST REPORT**HP MILJUTE 460 DRUM (210 L)(1370241)**

Laboratory Serial No.	FB 1892	Sample Drawn By	OPN OFFICER
JDE Serial No	1226747	Date Sample Drawn	08/03/23
Batch No / Lot No	M2-23-C-447	Date Received In Lab	08/03/23
Work Order No	1030034	Date Tested	08/03/23
Tank /Kettle/Location	WHS2	Particular of Sample	DRUM
Plant Code/ Name		Sampling Plan	IS 1447

Test Line Number	Test Description/ Characteristics	Units	Test Method	Specifications		Result Value
				Min	Max	
1	APPEARANCE	.	VISUAL	B&C	B&C	B&C
2	KV @ 40°C	cSt	P25 D445 D7042	420.00	500.00	495.30
3	DENSITY@29.5°C	gm/cc	P16 D1298 D4052	-	-	.8959
4	FLASH POINT	°C	P69 D92	240	-	318

Remarks : Sample meets the specification in above tests.

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LABORATORY TEST REPORT**TRIMOFIN 55 DRUM 210 LIT(0127241)**

Laboratory Serial No.	PP - 011	Sample Drawn By	QC OFFICER
JDE Serial No	1234216	Date Sample Drawn	04/04/23
Batch No / Lot No	M1-23-D-509	Date Received In Lab	04/04/23
Work Order No	1039121	Date Tested	05/04/23
Tank /Kettle/Location	WHS	Particular of Sample	DRUM
Plant Code/ Name		Sampling Plan	IS 1447

Test Line Number	Test Description/ Characteristics	Units	Test Method	Specifications		Result Value
				Min	Max	
1	TRIMOFIN 55 COLOUR	ASTM	P 12 D1500 IP196	-	8.0	4.5
2	TRIMOFIN 55 APPEARANCE	.	VISUAL	B&C	B&C	B&C
3	TRIMOFIN 55 CHLORINE	%Wt	P 50 D 1317 D 6443 IP 118	1.6	2.0	1.8
4	TRIMOFIN 55 K V @ 40	cSt	P 25 D 7042 D 445 IP 71	32	42	36
5	TRIMOFIN 55 DENSITY@29.5	gm/cc	P 16 D 1298 IP 160 D 4052	-	-	.894
6	TRIMOFIN 55 FLASH POINT	COC	P 16 D 1298 IP 160	160	-	196

Remarks : Sample meets the specification in above tests.

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DALVI P Y(30044160)

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MUMBAI,400033

LABORATORY TEST REPORT**ENKLO 32 (210 L)(0202241)**

Laboratory Serial No.	FB 0074	Sample Drawn By	OPN OFFICER
JDE Serial No	1236601	Date Sample Drawn	17/04/23
Batch No / Lot No	M2-23-D-713	Date Received In Lab	17/04/23
Work Order No	1041759	Date Tested	17/04/23
Tank /Kettle/Location	WHS2	Particular of Sample	DRUM
Plant Code/ Name	14546	Mazgaon TML Lubes-(STP 0101)	Sampling Plan
			IS 1447

Test Line Number	Test Description/ Characteristics	Units	Test Method	Specifications		Result Value
				Min	Max	
16	ZINC	%Wt	ASTMD6443 ,EDTA	.026	.070	.034
1	APPEARANCE	Visual	VISUAL	B&C	B&C	B&C
2	DENSITY@29.5°C	gm/cc	P16, ASTMD1298,D4052,D7042	-	-	.8710
3	EMUL CHAR@54°C /20MIN	Milliliter	IS1448(P91),ASTM D1401	40-40-0	40-37-3	40-39-1
4	FLASH POINT, COC°C	°C	IS 1448(P:69), ASTM D 92	190	-	220
5	FOAMING STABILITY SEQ1	Milliliter	IS 1448(P:67), ASTM D892	0	0	0
6	FOAMING STABILITY SEQ2	Milliliter	IS 1448(P:67), ASTM D892	0	0	0
7	FOAMING STABILITY SEQ3	Milliliter	IS 1448(P:67), ASTM D892	0	0	0
8	FOAMING TENDENCY SEQ1	Milliliter	IS 1448(P:67), ASTM D892	0	50	0
9	FOAMING TENDENCY SEQ2	Milliliter	IS 1448(P:67), ASTM D892	0	50	20
10	FOAMING TENDENCY SEQ3	Milliliter	IS 1448(P:67), ASTM D892	0	50	20
11	K V @ 100° C	cSt	IS 1448(P25),ASTM D7042 ,D 445	-	999999999.900	5.381
12	K V @ 40° C	cSt	IS 1448(P25),ASTM D7042 ,D 445	29.00	34.00	33.09
13	VISCOSITY INDEX	.	IS 1448(P56),ASTM D2270	90	-	94
14	POUR POINT°C	°C	IS 1448(P10),ASTM D97	-	-3	-15
15	RUSTTEST,SYNTHSEAWATER	.	IS 1448(P96),ASTM D665	P	P	P
17	ENKLO 32 COPSTRIPCORR@100	.		-	1	1

Remarks : Sample meets the specification in above tests.

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Reviewed and Authorized by:
PARVEEN RUBANA(31989940)

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QUALITY CONTROL LABORATORY

MAZGAON TERMINAL

LUBE PLANT, HAY BUNDER ROAD, MAZGAON

MUMBAI,400033

LABORATORY TEST REPORT**HP MILCY NO. 1 PLUS (210 L)(2108241)**

Laboratory Serial No.	FB--0098	Sample Drawn By	OPN OFFICER
JDE Serial No	1237368	Date Sample Drawn	20/04/23
Batch No / Lot No	M1-23-D-573	Date Received In Lab	20/04/23
Work Order No	1043379	Date Tested	20/04/23
Tank /Kettle/Location	WHS	Particular of Sample	DRUM
Plant Code/ Name		Sampling Plan	IS 1447

Test Line Number	Test Description/ Characteristics	Units	Test Method	Specifications		Result Value
				Min	Max	
1	APPEARANCE	.	VISUAL	B&C	B&C	B&C
2	CCS @ -20°C	Centi Poise	IS1448(P108),ASTMD5293	-	7000	6559
3	DENSITY@29.5°C	gm/cc	P16 D4052 D1298 D7042	-	-	.8730
4	FLASHPOINT,COC	°C	IS1448(P69), ASTMD92	214	-	234
5	SULFATED ASH	%Wt	IS1448(P4), ASTMD874	.992	1.650	1.370
6	TBN	mg KOH/g	IS1448(P86),ASTMD2896	9.90	-	10.51
7	ZINC	%Wt	ASTMD6443, EDTA	.089	.126	.106
8	SULFATED ASH	%Wt	IS1448 (P4),ASTMD874	1.210	1.490	1.370
9	KV @ 100°C	cSt	IS1448(P25),ASTMD445,D7042	14.00	15.00	14.20
10	POUR POINT	°C	IS1448 (P10), ASTMD97	-	-24	-30
11	VISCOSITY INDEX (VI)	.	IS1448(P56), ASTM D2270	120	-	134

Remarks : SAMPLE MEETS THE SPECIFICATION IN ALL ABOVE TEST PARAMETERS.

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Reviewed and Authorized by:

WARANG CHANDRAKANT PRABHAKAR(30074520)

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MUMBAI,400033

LABORATORY TEST REPORT**WAYLUBE N-68 (210 L)(0360241)**

Laboratory Serial No.	PP - 02	Sample Drawn By	QC OFFICER
JDE Serial No	1237385	Date Sample Drawn	01/04/23
Batch No / Lot No	M1-23-C-483	Date Received In Lab	01/04/23
Work Order No	1041215	Date Tested	01/04/23
Tank /Kettle/Location	WHS	Particular of Sample	DRUM
Plant Code/ Name	14546	Mazgaon TML Lubes-(STP 0101)	Sampling Plan
			IS 1447

Test Line Number	Test Description/ Characteristics	Units	Test Method	Specifications		Result Value
				Min	Max	
1	WAYLUBE N 68 APPEARANCE	.	VISUAL	C	C	C
3	WAYLUBE N 68 DENSITY@29.5	gm/cc	P 16 D 1298 IP 160	-	-	.878
4	WAYLUBE N 68 FLASH POINT	COC	P 69 D 92 IP 36	190	-	192
8	WAYLUBE N 68 KV @ 40	cSt	P 25 D 445 IP 71	62	68	64
7	WAYLUBE N 68 SAPNIFICATION VAL	mg KOH/g	P 55 D 94	8.0	11.0	10.8
9	WAYLUBE N 68 VI	.	P 56 D 2270 IP 226	100	-	113
2	WAYLUBE N 68 COR CU STRIP@100	ASTM	P 15 D 130 IP 154	-	1	1
6	WAYLUB N68 RUST TESTD/W	.	P96 D665 IP135	P	P	P
5	WAYLUBEN68 NEUTRALISATION NUMB	mg KOH/g	P 2 D 974 IP 77	-	.25	.20

Remarks : Sample meets the specification in above tests.

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Reviewed and Authorized by:
DALVI P Y(30044160)

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LABORATORY TEST REPORT**SPINTEK 12 (210 L)(0657241)**

Laboratory Serial No.	PP-94	Sample Drawn By	QC OFFICER
JDE Serial No	1238840	Date Sample Drawn	24/04/23
Batch No / Lot No	M1-23-D-703	Date Received In Lab	24/04/23
Work Order No	1044460	Date Tested	24/04/23
Tank /Kettle/Location	WHS	Particular of Sample	DRUM
Plant Code/ Name	14546	Mazgaon TML Lubes-(STP 0101)	Sampling Plan
			IS 1447

Test Line Number	Test Description/ Characteristics	Units	Test Method	Specifications		Result Value
				Min	Max	
2	SPINTEK 12 DENSITY@29.5	gm/cc	P 16 D 1298 IP 160	-	-	.8515
4	SPINTEK 12 KV @ 40	cSt	P 25 D 445 IP 71	9.00	13.00	12.65
5	SPINTEK 12 VI	.	P 56 D 2270 IP 226	90	-	90
6	SPINTEK 12 ACIDITY TOTAL	mg KOH/g	P 1 D 974 IP 139	-	.20	.15
1	SPINTEK 12 APPEARANCE	.	VISUAL	B&C	B&C	B&C
3	SPEINTEK 12 RUST TEST D/W	.	P 96 D665 IP 135	P	P	P

Remarks : SAMPLE MEETS THE SPECIFICATION IN ALL ABOVE TEST PARAMETERS.

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LABORATORY TEST REPORT**KOOLKUT 40 DRUM (210 L)(0503241)**

Laboratory Serial No.	FB-188	Sample Drawn By	OPERATION OFFICER
JDE Serial No	1241786	Date Sample Drawn	12/05/23
Batch No / Lot No	M1-23-E-763	Date Received In Lab	12/05/23
Work Order No	1049386	Date Tested	12/05/23
Tank /Kettle/Location	WHS	Particular of Sample	DRUM
Plant Code/ Name		Sampling Plan	IS 1447

Test Line Number	Test Description/ Characteristics	Units	Test Method	Specifications		Result Value
				Min	Max	
4	APPEARANCE	.	VISUAL	C	C	C
4	EMULSION TEST	.	IS 1115	P	P	P
4	KV@40°C	cSt	IS1448(P25), ASTM D445,D7042	20.00	-	29.79
4	THERMAL STABILITY	.	IS 1115	P	P	P

Remarks : SAMPLE MEETS THE SPECIFICATION IN ALL ABOVE TEST PARAMETERTS.

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MUMBAI,400033

LABORATORY TEST REPORT**KOOLKUT 40 DRUM (210 L)(0503241)**

Laboratory Serial No.	FB - 209	Sample Drawn By	QC OFFICER
JDE Serial No	1242607	Date Sample Drawn	18/05/23
Batch No / Lot No	M1-23-E-776	Date Received In Lab	18/05/23
Work Order No	1050757	Date Tested	19/05/23
Tank /Kettle/Location	WHS	Particular of Sample	DRUM
Plant Code/ Name		Sampling Plan	IS 1447

Test Line Number	Test Description/ Characteristics	Units	Test Method	Specifications		Result Value
				Min	Max	
4	APPEARANCE	.	VISUAL	C	C	C
4	EMULSION TEST	.	IS 1115	P	P	P
4	KV@40°C	cSt	IS1448(P25), ASTM D445,D7042	20.00	-	27.20
4	THERMAL STABILITY	.	IS 1115	P	P	P

Remarks : Sample meets the specification in above tests.

Note:

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LABORATORY TEST REPORT**TRANSFORMER OIL 210L EC DRUM(2376211)**

Laboratory Serial No.	FB-110	Sample Drawn By	OPERATION OFFICER
JDE Serial No	1243556	Date Sample Drawn	22/04/23
Batch No / Lot No	M1-23-D-447	Date Received In Lab	22/04/23
Work Order No	1051705	Date Tested	24/04/23
Tank /Kettle/Location	WHS	Particular of Sample	DRUM
Plant Code/ Name		Sampling Plan	IS :6855

Test Line Number	Test Description/ Characteristics	Units	Test Method	Specifications		Result Value
				Min	Max	
1	TRANSFORMER OIL FLASH POINT	°C	IS 1448 P 21	135.0	300.0	168.4
2	TRANSFORMER OIL FURFURL	mg/kg	IS 15668	0	.05	0
3	TRANSFORMER OIL GAS TEND	mm3/min	IEC 60628 METHOD A	-	9999	9.8
4	TRANSFORMER OIL ACID OXIDISED	mg KOH/g	12422 (C) 164 h 1.9.4	0	1.20	.06
5	TRANSFORMER OIL DDF OXIDISED	.	IS 12422(C) 164 H, 1.9.6	0	.500	.002
6	TRANSFORMER OIL SLUD OXIDISED	wt. %	IS 12422(C) 164 H, 1.9.1	0	.80	.06
7	TRANSFORMER OIL ACIDITY	mg KOH/g	IEC 62021-1	0	.01	0
8	TRANSFORMER OIL APPEARANCE	.	VISUAL	B&C	B&C	B&C
9	TRANSFORMER OIL BDV	kV	IS 6792	30	999	72
10	TRANSFORMER OIL CORROSI SULFUR	.	DIN 51353	NOT CORROSIVE	NOT CORROSIVE	NOT CORROSIVE
11	TRANSFORMER OIL DBDS	mg/kg	IS16497 PART 1	0	5.000	0
12	TRANSFORMER OIL DDF@90	.	IS 16086	0	.005000	.001400
13	TRANSFORMER OIL DENSITY@20	g/ml	IS 1448 P 16	.8000	.8950	.8401
14	TRANSFORMER OIL INHIBITOR	wt. %	IS 13631	0	.010	.010
15	TRANSFORMER OIL INTER TENSION	mN/m	ASTM D 971	-	-	42.0
16	TRANSFORMER OIL KV@0°C	mm2/sec	IS 1448 P 25	0	1800.00	66.74
17	TRANSFORMER OIL KV@40°C	mm2/sec	IS 1448 P 25	0	15.00	11.27
18	TRANSFORMER OIL METAL PASSIVAT	mg/kg	IS 13631	1.00	5.00	1.00
19	TRANSFORMER OIL PARTICLE COUNT	.	IS 13236	-	10000000000000	5
20	TRANSFORMER OIL PCA	wt. %	IP 346	0	3.00	1.00
21	TRANSFORMER OIL PCB	mg/kg	IS 16082	0	2	0
22	TRANSFORMER OIL POT COR SULFUR	.	IS 16310	NOT CORROSIVE	NOT CORROSIVE	NOT CORROSIVE
23	TRANSFORMER OIL POUR POINT	°C	IS 1448 P 10	-99	-10	-12
24	TRANSFORMER OIL TOTAL SULFUR	wt. %	ISO 14596 OR ASTM D 4294	0	99999998.951	0
25	TRANSFORMER OIL WATER CONTENT	mg/kg	IEC 60814 NOTE 3 IS 335:2018	0	30	19

Remarks : SAMPLE MEETS THE SPECIFICATION IS 335 : 2018 IN ALL ABOVE TEST PARAMETERS.

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MUMBAI,400033

LABORATORY TEST REPORT**PARTHAN EP 320 (210 L)(0883241)**

Laboratory Serial No.	FB-242	Sample Drawn By	OPERATION OFFICER
JDE Serial No	1244867	Date Sample Drawn	27/05/23
Batch No / Lot No	M1-23-E-972	Date Received In Lab	27/05/23
Work Order No	1054295	Date Tested	28/05/23
Tank /Kettle/Location	WHS	Particular of Sample	DRUM
Plant Code/ Name	14546	Mazgaon TML Lubes-(STP 0101)	Sampling Plan
			IS 1447

Test Line Number	Test Description/ Characteristics	Units	Test Method	Specifications		Result Value
				Min	Max	
1	APPEARANCE	.	VISUAL	C	C	C
2	ASH	%Wt	IS 1448(P4), ASTMD482	-	.100	.010
3	DENSITY@29.5°C	gm/cc	P16,ASTMD4052, D1298, D7042	-	-	.8669
4	FLASH POINT°C,COC	°C	IS1448(P69),ASTMD92	220	-	228
6	FOAMING TEST SEQ1	Milliliter	IS1448(P67), ASTMD892		100/0	0/0
7	FOAMING TEST SEQ2	Milliliter	IS1448(P67), ASTMD892	-	50/0	10/0
8	FOAMING TEST SEQ3	Milliliter	IS1448(P67), ASTMD892	-	100/0	0/0
9	RUST TEST A/B	.	IS1448(P96),ASTMD665	P	P	P
10	KINEMATIC VISCOSITY @40°C	cSt	IS1448(P25),ASTMD445, D7042	300.0	340.0	323.2
5	EMULSION TEST@82°C/30MIN	Milliliter	IS1448(P91), ASTMD1401	40-40-0	40-40-0	40-40-0
11	PHOSPHORUS	%Wt	IS1448(P101),ASTMD1091, D6443	.0098	.0145	.0099

Remarks : SAMPLE MEETS THE SPECIFICATION IN ALL ABOVE TEST PARAMETERS.

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MUMBAI,400033

LABORATORY TEST REPORT**HP MILCY POWER 15W 40 (210L)(0818241)**

Laboratory Serial No.	FB-309	Sample Drawn By	OPERATION OFFICER
JDE Serial No	1247614	Date Sample Drawn	10/06/23
Batch No / Lot No	M1-23-F-102	Date Received In Lab	10/06/23
Work Order No	1057769	Date Tested	11/06/23
Tank /Kettle/Location	WHS	Particular of Sample	DRUM
Plant Code/ Name	14546	Mazgaon TML Lubes-(STP 0101)	Sampling Plan
			IS 1447

Test Line Number	Test Description/ Characteristics	Units	Test Method	Specifications		Result Value
				Min	Max	
1	APPEARANCE	.	VISUAL	B&C	B&C	B&C
8	CCS @ -20°C	Centi Poise	P108 D5293	-	7000	5079
2	DENSITY@29.5°C	gm/cc	P16 D1298 D4052 D7042	-	-	.8812
3	FLASHPOINT	°C	P69 D92	190	-	226
6	KV @ 100°C	cSt	P25 D445 D7042	14.00	15.20	14.23
5	KV @ 40°C	cSt	P25 D445 D7042	-	-	102.9
4	POUR POINT	°C	P 10 D 97	-	-21	-24
7	VISCOSITY INDEX (VI)	.	P56 D2270	120	-	141
9	CALCIUM	%Wt	ASTMD6443,EDTA	.391	.478	.401
10	SULFATED ASH	%Wt	P 4 D 874 IP 163	1.50	1.83	1.61
11	TBN	mg KOH/g	P86 D2896	11.3	-	12.3
12	ZINC	%Wt	ASTMD6443,EDTA	.112	.137	.116

Remarks : SAMPLE MEETS THE SPECIFICATION IN ALL ABOVE TEST PARAMETERS.

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LUBE PLANT, HAY BUNDER ROAD, MAZGAON

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LABORATORY TEST REPORT**HP MILCY POWER 15W 40 (210L)(0818241)**

Laboratory Serial No.	FB-352	Sample Drawn By	OPERATION OFFICER
JDE Serial No	1249800	Date Sample Drawn	19/06/23
Batch No / Lot No	M1-23-F-134	Date Received In Lab	19/06/23
Work Order No	1060634	Date Tested	19/06/23
Tank /Kettle/Location	WHS	Particular of Sample	DRUM
Plant Code/ Name	14546	Mazgaon TML Lubes-(STP 0101)	Sampling Plan
			IS 1447

Test Line Number	Test Description/ Characteristics	Units	Test Method	Specifications		Result Value
				Min	Max	
1	APPEARANCE	.	VISUAL	B&C	B&C	B&C
7	CCS @ -20°C	Centi Poise	P108 D5293	-	7000	5418
2	FLASHPOINT	°C	P69 D92	190	-	220
5	KV @ 100°C	cSt	P25 D445 D7042	14.00	15.20	14.53
4	KV @ 40°C	cSt	P25 D445 D7042	-	-	106
3	POUR POINT	°C	P 10 D 97	-	-21	-30
6	VISCOSITY INDEX (VI)	.	P56 D2270	120	-	141
8	CALCIUM	%Wt	ASTMD6443,EDTA	.391	.478	.467
9	SULFATED ASH	%Wt	P 4 D 874 IP 163	1.50	1.83	1.74
10	TBN	mg KOH/g	P86 D2896	11.3	-	13.2
11	ZINC	%Wt	ASTMD6443,EDTA	.112	.137	.135

Remarks : SAMPLE MEETS THE SPECIFICATION IN ALL ABOVE TEST PARAMETERS.

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LABORATORY TEST REPORT**TRANSFORMER OIL 210L EC DRUM(2376211)**

Laboratory Serial No.	FB-221	Sample Drawn By	OPERATION OFFICER
JDE Serial No	1250310	Date Sample Drawn	22/05/23
Batch No / Lot No	M1-23-E-777	Date Received In Lab	22/05/23
Work Order No	1060984	Date Tested	25/05/23
Tank /Kettle/Location	WHS	Particular of Sample	DRUM
Plant Code/ Name		Sampling Plan	IS 6855

Test Line Number	Test Description/ Characteristics	Units	Test Method	Specifications		Result Value
				Min	Max	
1	TRANSFORMER OIL FLASH POINT	°C	IS 1448 P 21	135.0	300.0	170.0
2	TRANSFORMER OIL FURFURL	mg/kg	IS 15668	0	.05	0
3	TRANSFORMER OIL GAS TEND	mm3/min	IEC 60628 METHOD A	-	9999	10.0
4	TRANSFORMER OIL ACID OXIDISED	mg KOH/g	12422 (C) 164 h 1.9.4	0	1.20	0
5	TRANSFORMER OIL DDF OXIDISED	.	IS 12422(C) 164 H, 1.9.6	0	.500	.002
6	TRANSFORMER OIL SLUD OXIDISED	wt. %	IS 12422(C) 164 H, 1.9.1	0	.80	.06
7	TRANSFORMER OIL ACIDITY	mg KOH/g	IEC 62021-1	0	.01	0
8	TRANSFORMER OIL APPEARANCE	.	VISUAL	B&C	B&C	B&C
9	TRANSFORMER OIL BDV	kV	IS 6792	30	999	72
10	TRANSFORMER OIL CORROSI SULFUR	.	DIN 51353	NOT CORROSIVE	NOT CORROSIVE	NOT CORROSIVE
11	TRANSFORMER OIL DBDS	mg/kg	IS16497 PART 1	0	5.000	0
12	TRANSFORMER OIL DDF@90	.	IS 16086	0	.005000	.001800
13	TRANSFORMER OIL DENSITY@20	g/ml	IS 1448 P 16	.8000	.8950	.8355
14	TRANSFORMER OIL INHIBITOR	wt. %	IS 13631	0	.010	.010
15	TRANSFORMER OIL INTER TENSION	mN/m	ASTM D 971	-	-	41.8
16	TRANSFORMER OIL KV@0°C	mm2/sec	IS 1448 P 25	0	1800.00	53.76
17	TRANSFORMER OIL KV@40°C	mm2/sec	IS 1448 P 25	0	15.00	10.06
18	TRANSFORMER OIL METAL PASSIVAT	mg/kg	IS 13631	1.00	5.00	1.00
19	TRANSFORMER OIL PARTICLE COUNT	.	IS 13236	-	10000000000000	5
20	TRANSFORMER OIL PCA	wt. %	IP 346	0	3.00	1.00
21	TRANSFORMER OIL PCB	mg/kg	IS 16082	0	2	0
22	TRANSFORMER OIL POT COR SULFUR	.	IS 16310	NOT CORROSIVE	NOT CORROSIVE	NOT CORROSIVE
23	TRANSFORMER OIL POUR POINT	°C	IS 1448 P 10	-99	-10	-12
24	TRANSFORMER OIL TOTAL SULFUR	wt. %	ISO 14596 OR ASTM D 4294	0	99999998.951	.001
25	TRANSFORMER OIL WATER CONTENT	mg/kg	IEC 60814 NOTE 3 IS 335:2018	0	30	16

Remarks : SAMPLE MEETS THE SPECIFICATION IS 335 : 2018 IN ALL ABOVE TEST PARAMERERS.

Note:

- 1) The report refers only to the sample tested.
- 2) The test report shall not be reproduced except in full without prior written approval of the Lab Incharge.
- 3) This test report shall not be used in any advertising media or as evidence in the Court of Law without prior written consent of Laboratory.
- 4) Test results reported are valid at the time of testing
- 5) This is a system generated test report and hence does not require signature.

Reviewed and Authorized by:

WARANG CHANDRAKANT PRABHAKAR(30074520)